ACTIVITY 6

Group Test Summary

Database Backup, Restoration, and Recovery

Prepared from the eight project folders in the class Group 5 repository and the supplied COMP 018 Activity 6 instructions.

Review basis: Part A–E Markdown reports for Groups 02, 03, 04, 06, 07, 08, 10 and 11, plus the Activity 6 assignment PDF uploaded in this conversation.

Overall result

All eight reports describe successful backup creation, restoration to separate test databases, verification, and recovery of the intended disposable test record. However, each report also documents limitations that prevent treating the work as an unqualified production-ready success. For this summary, each group is classified as SOME ISSUES.

Group	Project	Restore outcome in report	Summary classification
02	Park It	Restored and verified; disposable vehicle recovered	SOME ISSUES
03	OJT Management System	Restored and verified; company fixture recovered	SOME ISSUES
04	BrewPOS	Restored and verified; menu-category fixture recovered	SOME ISSUES
06	HabalGo	Restored and verified; vehicle fixture recovered	SOME ISSUES
07	CampusSafe	Restored and verified; hazard-category fixture recovered	SOME ISSUES
08	Campus Canteen	Restored and verified; menu-item fixture recovered	SOME ISSUES
10	MESA	Restored and verified; menu-item fixture recovered	SOME ISSUES
11	BoardTrack	Restored and verified; property fixture recovered	SOME ISSUES

Important interpretation

These are summaries of what the repository reports claim and document. The tests were not independently rerun for this PDF. The reports themselves state that AI-assisted commands were validated through PostgreSQL outputs and comparison checks, and that no human validation is claimed.

Prepared 6 November 2026 • Source repository: grp05-dba/grp05_lostandfound • Branch: main

1. What the assignment requires

The uploaded Activity 6 instructions require a project-by-project progress document. The required content is mapped below to the group reports reviewed.

Required section	What should be shown	What the repository reports contain
Projects covered	Project, repository, database and accounts	Project names, source database names, backup/restore accounts and test-target names are documented across Parts A–E.
Backup plan	Objects, method, storage/access, verification, schedule, retention, RPO/RTO, role recreation	All groups describe PostgreSQL logical custom-format backups, local protected storage, verification, proposed nightly schedule, retention and recovery objectives.
Backup creation & inspection	Commands, file name, size, checksum, archive listing	Each report says original and fixture archives were created, inspected and checksummed. Exact metadata varies by group; Group 02 includes exact sizes and hashes in Part B.
Restoration & verification	Separate test DB; tables, records, constraints and queries	Each group documents separate restore targets, row/schema comparisons, object and constraint checks, and representative reads.
Access control	Activity 4 roles/memberships and allowed/denied tests	Groups report successful permitted reads and denied reporting writes; special restore authority phases are documented where needed.
Recovery simulation	Incident, timeline, method, expected-vs-recovered state	Each group created a disposable fixture, verified a snapshot before deletion, deleted only the test fixture, and restored it to a separate recovered database.
Evaluation & recommendations	Real-world suitability and readiness improvements	Nightly/24-hour RPO/60-minute RTO is treated as a classroom baseline, not a guaranteed fit for real systems.
Limitations & errors	Warnings, permission limits, errors and handling	The reports discuss sample-data limits, role migration not tested on fresh clusters, environment limitations, and dump/reparse/probe adjustments.
Feedback & AI declaration	Owner feedback status; AI use and validation	Reports declare AI assistance. Owner feedback, tracker update and final PDF submission are generally stated as not yet completed.

Shared backup policy proposed by the reports

Method: PostgreSQL logical backup using `pg_dump -Fc`, inspected with `pg_restore --list` and restored with `pg_restore`. **Proposed schedule:** nightly, typically 02:00 Asia/Manila, plus before major database changes. **Retention:** seven daily, four weekly and three monthly points. **Proposed objectives:** 24-hour RPO and 60-minute RTO. These are proposals; the reports say scheduled jobs, offsite/encrypted copies and retention automation were not installed as operational services.

The assignment also says data-bearing backup files must remain outside Git and all destructive tests must be restricted to disposable test databases. The reports state that original databases and archives were preserved.

2. Group 02 — Park It

SOME ISSUES	**Source database:** `grp_parkitdb` **Backup/restore account:** `parkit_dba`

Test environment: The report describes a local PostgreSQL 18.6 test on port 55442 against the referenced Park-It database.

Backup creation and inspection

Two custom-format archives were created: the original database and a fixture backup from the restored test database. Both commands reportedly exited 0; both archives were listed with pg_restore --list and SHA-256 checked.

Restoration and verification

Restored to parkit_a6_restore. Five tables, five sequences and nine indexes were present; row counts matched (1 parking zone, 2 staff users, and initially empty vehicle/transaction/payment tables). Constraints, ownership, sequence state, and effective permissions were checked.

Access-control and application checks

Runtime SELECT and auditor approved-column SELECT succeeded. Auditor UPDATE and password_hash access were denied. PHP/PDO connected as parkit_app; staff login, dashboard, monitoring and logout succeeded.

Controlled recovery simulation

A disposable vehicle ACT6-RECOVER-02 (ID 12) was included in a separately verified fixture backup, then deleted only from parkit_a6_restore. It was restored exactly into parkit_a6_recovered. A later vehicle was intentionally absent from the recovered snapshot.

Group-specific notes

Two owner-only sequence ACLs appeared as default NULL ACLs after restore, but effective permissions were equivalent. The six Activity 4 roles and three memberships were verified in the existing cluster; fresh-cluster role recreation was not tested.

Limitations

Operational tables were mostly empty; no realistic parking/payment workload, production RTO, PITR, or production cutover. Existing application authorization/business-rule issues remain. Backups were in /tmp on the same host; proposed durable/offsite retention was not implemented.

Separate restore/recovery targets: `parkit_a6_restore / parkit_a6_recovered`
Disposable fixture: ACT6-RECOVER-02 vehicle row

2. Group 03 — OJT Management System

SOME ISSUES	**Source database:** `ojt_managementdb` **Backup/restore account:** `grp03_ojt_dba`

Test environment: The original referenced Activity 4 environment was unavailable on the test PC; the report says a documented local copy was reconstructed from reviewed setup and role sources.

Backup creation and inspection

Custom-format original and fixture archives were created, inspected, sized and SHA-256 checked. Data-bearing archives stayed outside Git.

Restoration and verification

Restored to g05_a6_g03_restore using an authority-aware restore phase. Matching row counts and whole-row fingerprints were reported across tables; schema, constraints, indexes, views, routines, triggers, sequences, role attributes and effective privileges were compared.

Access-control and application checks

Runtime/reporting read query succeeded; reporting UPDATE was denied (SQLSTATE 42501). Java JDBC application authenticated five demo roles and built initial tabs headlessly on restore and recovered targets; full interactive workflows were not tested.

Controlled recovery simulation

Disposable company fixture G05-A6-RECOVER-03 was backed up and independently restored to g05_a6_g03_snapshot before deletion. It was deleted only in the restore copy, then recovered into g05_a6_g03_recovered. A post-snapshot G05-A6-LATE-03 row was intentionally not in the recovered copy.

Group-specific notes

Restore used explicit DBA/owner authority; no separate login-default or extension ACL phase was required. The report notes narrow normalization of equivalent varchar-array casts and corrected invalid-enum probe bookkeeping before final checks.

Limitations

Fresh-cluster role recreation was supplied but not tested. Small sample dataset and empty workflow tables; no realistic production stream, production RTO benchmark, failover/cutover or PITR. Reconstructed environment is not the original owner's deployed instance.

Separate restore/recovery targets: `g05_a6_g03_restore / g05_a6_g03_recovered`
Disposable fixture: G05-A6-RECOVER-03 company row

2. Group 04 — BrewPOS

SOME ISSUES	**Source database:** `grp_brewposdb` **Backup/restore account:** `brewpos_dba`

Test environment: The original referenced Activity 4 environment was unavailable on the test PC; a documented local copy was reconstructed from reviewed setup and role sources.

Backup creation and inspection

Custom-format original and fixture archives were created, inspected, sized and SHA-256 checked.

Restoration and verification

Restored to g05_a6_g04_restore. The five tables (menu_category, menu_item, order_line_item, order_ticket, staff) had matching source/restore row counts. Schema, constraints, indexes, routines, triggers, sequences, ownership and effective permissions were compared.

Access-control and application checks

Runtime/reporting read query succeeded and reporting UPDATE was denied (42501). Spring Boot app returned HTTP 200 from /api/staff on restore and recovered targets; no checkout/write or full UI suite was run.

Controlled recovery simulation

Disposable menu category G05-A6-RECOVER-04 was snapshot-verified, deleted only from g05_a6_g04_restore, then recovered into g05_a6_g04_recovered. The post-snapshot G05-A6-LATE-04 marker was intentionally not recovered.

Group-specific notes

Restore preserved owners/grants and copied per-database search_path, statement_timeout and lock_timeout settings. A single restore phase was reported; no separate login-default or extension ACL phase was needed.

Limitations

Fresh-cluster migration, production cutover and cross-version restore were not tested. Small sample data; no production RTO benchmark, failover or PITR. Proposed schedule/retention/encryption were not operationalized. Existing Activity 4 issues remain.

Separate restore/recovery targets: `g05_a6_g04_restore / g05_a6_g04_recovered`
Disposable fixture: G05-A6-RECOVER-04 menu category

2. Group 06 — HabalGo

SOME ISSUES	**Source database:** `grp00_habalgodb` **Backup/restore account:** `habalgo_dba`

Test environment: The original referenced Activity 4 environment was unavailable on the test PC; a documented local copy was reconstructed from reviewed setup and role sources.

Backup creation and inspection

Custom-format original and fixture archives were created, inspected, sized and SHA-256 checked.

Restoration and verification

Restored to g05_a6_g06_restore. All eight listed tables had matching source/restore row counts (only driver contained 10 rows; the other seven were empty). Schema, constraints, indexes, routines, triggers, sequences and effective privileges were compared.

Access-control and application checks

Runtime/reporting read query succeeded; reporting UPDATE was denied (42501). PHP PDO database/configuration component connected and read both restore and recovered targets with the restricted runtime identity; this was not a full interactive web test.

Controlled recovery simulation

Disposable vehicle G05-A6-RECOVER-06 was snapshot-verified, deleted only from g05_a6_g06_restore, and recovered into g05_a6_g06_recovered. Post-snapshot G05-A6-LATE-06 was absent from the recovered snapshot as expected.

Group-specific notes

Authority-aware restore preserved owners and grants. Per-database connection/search_path/timeout policy was applied. The Part E report notes that fresh-cluster role recreation was not tested.

Limitations

The original referenced environment was unavailable and was reconstructed. Very sparse sample data; no realistic bookings/payments workload, production RTO benchmark, failover/cutover or PITR. Existing application limitations remain.

Separate restore/recovery targets: `g05_a6_g06_restore / g05_a6_g06_recovered`
Disposable fixture: G05-A6-RECOVER-06 vehicle row

2. Group 07 — CampusSafe

SOME ISSUES	**Source database:** `grp07_campussafedb` **Backup/restore account:** `campussafe_dba`

Test environment: The original referenced Activity 4 environment was unavailable on the test PC; a documented local copy was reconstructed from reviewed setup and role sources.

Backup creation and inspection

Custom-format original and fixture archives were created, inspected, sized and SHA-256 checked.

Restoration and verification

Restored to g05_a6_g07_restore. Nine tables had matching source/restore row counts; data included 16 hazard reports and 60 report-status-history rows. Whole-row fingerprints and schema, constraints, indexes, routines, triggers, sequences and effective privileges were compared.

Access-control and application checks

Runtime/reporting read query succeeded; reporting UPDATE was denied (42501). PHP PDO database/configuration component connected and read restore and recovered targets using the restricted runtime identity; no full interactive web workflow was claimed.

Controlled recovery simulation

Disposable hazard category G05-A6-RECOVER-07 was snapshot-verified, deleted only from g05_a6_g07_restore, then recovered into g05_a6_g07_recovered. Post-snapshot G05-A6-LATE-07 was absent from the recovered copy as expected.

Group-specific notes

Authority-aware restore; target CONNECT was limited to planned roles. Source per-database runtime search_path/timeout settings were copied to the target.

Limitations

Reconstructed environment, not the original owner's deployment. Fresh-cluster role migration, production RTO, failover/cutover and PITR were not tested. Existing application authorization/integrity limitations remain.

Separate restore/recovery targets: `g05_a6_g07_restore / g05_a6_g07_recovered`
Disposable fixture: G05-A6-RECOVER-07 hazard category

2. Group 08 — Campus Canteen Pre-Order and Queue Management System

SOME ISSUES	**Source database:** `canteen_g05_activity4` **Backup/restore account:** `canteen_admin`

Test environment: The report used an existing local Activity 4/5 evaluation instance; it explicitly does not claim this was the production owning-group deployment.

Backup creation and inspection

Custom-format original and fixture archives were created, inspected, sized and SHA-256 checked.

Restoration and verification

Restored to g05_a6_g08_restore. Seven tables had matching row counts (3 students, 4 vendors, 10 menu items, 2 orders, 2 order items, 2 queue entries, 3 notifications). Schema, constraints, indexes, routines, triggers, sequences and effective privileges were compared.

Access-control and application checks

Runtime/reporting read query succeeded; reporting UPDATE was denied (42501). Spring Boot app returned HTTP 200 from /api/vendors on restore and recovered targets; no checkout/write or full UI suite was run.

Controlled recovery simulation

Disposable menu item G05-A6-RECOVER-08 was snapshot-verified, deleted only from g05_a6_g08_restore, and recovered into g05_a6_g08_recovered. Post-snapshot G05-A6-LATE-08 was not in the recovered copy.

Group-specific notes

Restore used an additional login_defaults phase for the login-owned default ACL. Target access was restricted to planned roles; no separate bootstrap extension ACL phase was needed.

Limitations

Existing local evaluation instance, not production owning-group deployment. Fresh-cluster role recreation, production RTO, failover/cutover and PITR were not tested. Sample data only; proposed backup schedule and retention not operational.

Separate restore/recovery targets: `g05_a6_g08_restore / g05_a6_g08_recovered`
Disposable fixture: G05-A6-RECOVER-08 menu item

2. Group 10 — MESA – Restaurant Online Ordering and Table Reservation System

SOME ISSUES	**Source database:** `mesa_g05_activity4` **Backup/restore account:** `rors_dba`

Test environment: The report used an existing local Activity 4/5 evaluation instance; it explicitly does not claim this was the production owning-group deployment.

Backup creation and inspection

Custom-format original and fixture archives were created, inspected, sized and SHA-256 checked.

Restoration and verification

Restored to g05_a6_g10_restore. Eight tables had matching row counts (51 customers, 10 staff, 50 menu items, 20 restaurant tables, 60 orders, 167 order items, 50 reservations, 56 payments). Schema, constraints, indexes, routines, triggers, sequences and effective privileges were compared.

Access-control and application checks

Runtime/reporting read query succeeded; reporting UPDATE was denied (42501). PHP PDO application database/configuration component connected and read restore and recovered targets with the restricted runtime identity; full interactive UI was not claimed.

Controlled recovery simulation

Disposable menu item G05-A6-RECOVER-10 was snapshot-verified, deleted only from g05_a6_g10_restore, then recovered into g05_a6_g10_recovered. Post-snapshot G05-A6-LATE-10 was absent as expected.

Group-specific notes

Restore used separate objects, login_defaults and extension_acl phases; extension ACLs were applied by provisioning administrator, without granting superuser to the DBA. This is the most involved restore authority setup among the groups.

Limitations

Existing local evaluation instance, not production deployment. Fresh-cluster migration, production RTO, failover/cutover and PITR not tested. Proposed schedule/retention/encryption not installed; sample dataset only.

Separate restore/recovery targets: `g05_a6_g10_restore / g05_a6_g10_recovered`
Disposable fixture: G05-A6-RECOVER-10 menu item

2. Group 11 — BoardTrack

SOME ISSUES	**Source database:** `boardtrack_g05_activity4` **Backup/restore account:** `boardtrack_dba`

Test environment: The report used an existing local Activity 4/5 evaluation instance; it explicitly does not claim this was the production owning-group deployment.

Backup creation and inspection

Custom-format original and fixture archives were created, inspected, sized and SHA-256 checked.

Restoration and verification

Restored to g05_a6_g11_restore. Eight tables had matching source/restore row counts (2 properties, 6 rooms, 4 tenants, 4 leases, 6 payments, 4 utility bills, 3 app users, 1 schema migration). Schema, constraints, indexes, routines, triggers, sequences and effective privileges were compared.

Access-control and application checks

Runtime/reporting read query succeeded; reporting UPDATE was denied (42501). Express pg-pool database component connected and read restore and recovered targets using the restricted runtime identity; full interactive UI was not claimed.

Controlled recovery simulation

Disposable property G05-A6-RECOVER-11 was snapshot-verified, deleted only from g05_a6_g11_restore, then recovered into g05_a6_g11_recovered. Post-snapshot G05-A6-LATE-11 was absent from recovered snapshot as expected.

Group-specific notes

Restore used objects, login_defaults and extension_acl phases; provisioning administrator handled extension ACLs. DBA was not granted superuser or lab_admin membership.

Limitations

Existing local evaluation instance, not production deployment. Fresh-cluster migration, production RTO, failover/cutover and PITR not tested. Proposed schedule/retention/encryption not installed.

Separate restore/recovery targets: `g05_a6_g11_restore / g05_a6_g11_recovered`
Disposable fixture: G05-A6-RECOVER-11 property row

3. Cross-group findings

Common successful test outcomes

• Every Part E report describes a custom-format PostgreSQL backup, archive inspection, a separate restore database, verification, and a controlled recovery test.

• Each Part D report describes verifying a fixture backup in a separate snapshot database before deleting the disposable record.

• Each report says the intended fixture record was present in the recovered database and absent from the incident database after deletion.

• Each report says a later record inserted after the snapshot was absent from the recovered copy. This is an expected recovery-point limitation, not a failed test.

• Access tests generally showed permitted runtime/reporting reads succeeding and reporting writes being denied with SQLSTATE 42501.

• Reports state that original databases and data-bearing backup files were preserved outside the repository.

Important differences between groups

Group	Notable detail
02	Park It includes explicit PHP login/dashboard/monitoring/logout checks, auditor denial of UPDATE and password-hash reads, and exact archive sizes/checksums. Its source operational tables were largely empty.
03	OJT Management System used a reconstructed local copy because the referenced Activity 4 environment was unavailable. The Java JDBC app authenticated five demo roles and built initial tabs headlessly; full interactive workflows were not tested.
04	BrewPOS used a reconstructed local copy. Spring Boot /api/staff returned HTTP 200, but checkout/write and full UI testing were not performed.
06	HabalGo used a reconstructed local copy. Seven of eight tables were empty; PHP PDO component connection/read checks were performed rather than a full interactive web test.
07	CampusSafe used a reconstructed local copy. Its sample data included 16 hazard reports and 60 status-history rows; PHP PDO component checks were reported.
08	Campus Canteen used a local Activity 4/5 evaluation instance, not the owning group's production deployment. Spring Boot /api/vendors returned HTTP 200; checkout/write tests were not run.
10	MESA used a local Activity 4/5 evaluation instance, not production. Its restore included a larger sample set, including 60 orders, 167 order items, 50 reservations and 56 payments.
11	BoardTrack used a local Activity 4/5 evaluation instance, not production. Restore included property, room, tenant, lease, payment and utility-bill records.

Restore authority notes

Groups 08, 10 and 11 document additional login-default restoration; Groups 10 and 11 also document a separate extension-ACL phase handled by a provisioning administrator. The reports emphasize that each phase was transactional, but the multi-phase restore as a whole was not one atomic transaction. Other groups generally used a single objects/owner restore phase.

4. Overall assessment and remaining gaps

Summary classification: SOME ISSUES for all eight groups

The classification reflects successful documented tests with qualifications. It does not mean the backup/restore/recovery operation failed. No group is classified as FAILED because every Part E report states that the intended record was restored and verified. No group is classified as unqualified SUCCESS because the reports explicitly identify limitations that matter for real deployment.

Area	What the reports establish	What remains unproven or incomplete
Backup/restore	Custom archives created, inspected and restored to separate test databases.	Fresh-cluster restoration and cluster-wide role/password recreation were not independently tested.
Data verification	Table counts, whole-row fingerprints, schema objects, constraints, sequences and permissions were compared.	Datasets are small or sparse; empty tables cannot demonstrate realistic transactional recovery.
Application checks	Restricted application identity connected; some endpoint or component checks succeeded.	Several groups did not run full UI workflows, writes/checkout, production cutover or end-to-end user journeys.
Recovery objectives	A 24-hour RPO and 60-minute RTO were proposed.	No production RTO benchmark, service failover or PITR/WAL recovery was demonstrated.
Storage/retention	Archives were kept outside Git in protected local directories.	The reports say offsite/encrypted copies, automated schedules and retention policies were proposed, not implemented; some storage was temporary/local.
Environment fidelity	Group 02 reports a local test against its project database; others report either reconstructed copies or local evaluation instances.	For Groups 03/04/06/07 the original referenced environment was unavailable; Groups 08/10/11 explicitly used local evaluation instances rather than production owner deployments.
Submission process	Part A–E reports and supporting metadata/evidence are in the shared Group 5 repository.	Reports say owner feedback was prepared but not sent, tracker updates were not completed, and final PDF/class submission was not done.

Recommended next steps before treating this as production-ready

• Re-run restore drills on an approved environment that faithfully represents each owning group's deployment, including a documented fresh-cluster role and membership recreation.

• Implement durable independent backup storage, encryption, monitoring, scheduled execution and retention; do not rely on temporary directories or same-host-only copies.

• Set RPO/RTO with each system owner. For payment, ordering, booking and operational records, assess whether a 24-hour RPO is acceptable; consider more frequent backups and WAL archiving/PITR where required.

• Test application reconnection and representative end-to-end workflows after restore, including safe writes where appropriate, then measure the full recovery time rather than only pg_restore duration.

• Send the documented findings to each owning group and complete the Activity 6 Log and PDF/tracker submission required by the assignment.

5. Source and scope notes

Assignment basis: COMP018_Activity_6.docx-1.pdf uploaded in this conversation, especially the PDF progress-documentation section and required deliverables (pages 4–6).

Repository reviewed: group-test-results/activity-6 on branch main. Each group folder's Part A–E Markdown files were used; Part C and Part D were reviewed for restoration/access and recovery details, and Part E for evaluation, limitations and AI-use declarations.

Group folder links:

Group 02 — Park It

Group 03 — OJT Management System

Group 04 — BrewPOS

Group 06 — HabalGo

Group 07 — CampusSafe

Group 08 — Campus Canteen Pre-Order and Queue Management System

Group 10 — MESA – Restaurant Online Ordering and Table Reservation System

Group 11 — BoardTrack

Scope limitation: This PDF summarizes the repository's submitted reports. It does not independently execute PostgreSQL commands, validate live database states, inspect the contents of every binary evidence attachment, or certify the groups' claims. Where a report states that a test was not performed, this summary preserves that limitation.